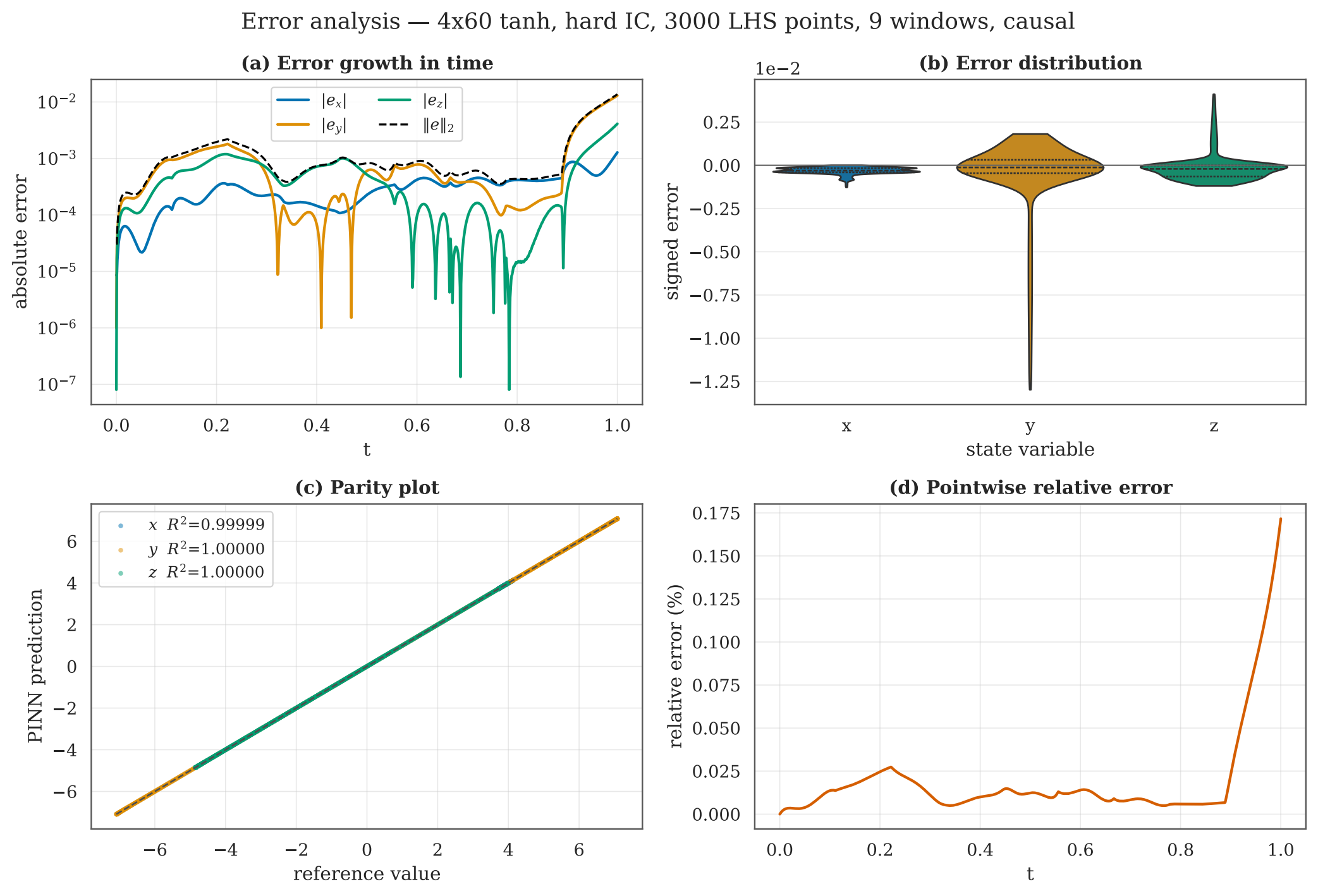

Error analysis — 4x60 tanh, hard IC, 3000 LHS points, 9 windows, causal
(a) Error growth in time
absolute error
t
|e_x|
|e_y|
|e_z|
‖e‖2
(b) Error distribution
1e−2
signed error
state variable
x
y
z
(c) Parity plot
PINN prediction
reference value
x R²=0.99999
y R²=1.00000
z R²=1.00000
(d) Pointwise relative error
relative error (%)
t